

GCSE Mathematics

Predicted Paper 3 — Higher Tier

Set A

Pearson Edexcel style (1MA1 / 3H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. Anya invests £3,200 in a savings account that pays simple interest at a rate of 4% per year. How much interest will she earn after 5 years?

(3 marks)

2. In a school, $\frac{3}{8}$ of the students wear glasses. Of those who wear glasses, $\frac{2}{5}$ also wear contact lenses. What fraction of all students wear both glasses and contact lenses? Give your answer in its simplest form.

(3 marks)

3. The formula for the volume of a sphere is $V = \frac{4}{3}\pi r^3$. A sphere has volume 500 cm^3 . Calculate its radius. Give your answer correct to 3 significant figures.

(3 marks)

4. n is an integer. Find all the integer values of n that satisfy $-3 < 2n + 1 \leq 7$.

(3 marks)

5. A car travels at an average speed of 72 km/h.

(a) Convert this speed into metres per second (m/s).

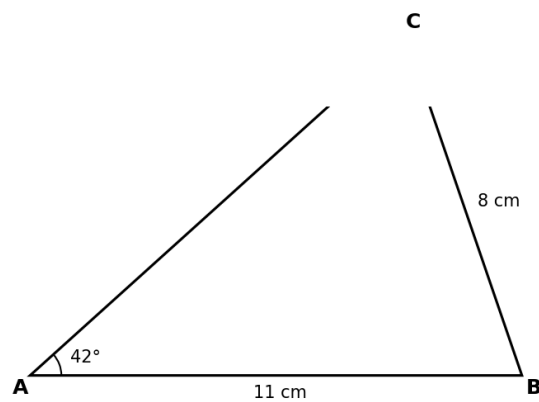
(2 marks)

(b) At this speed, how far does the car travel in 45 seconds? Give your answer in metres.

(2 marks)

6. In triangle ABC, AB = 11 cm, BC = 8 cm and angle BAC = 42° . Work out the size of angle BCA. Give your answer correct to 1 decimal place.

(4 marks)



7. The table shows the heights, in cm, of 50 plants in a garden.

Height, h (cm)	$0 < h \leq 10$	$10 < h \leq 20$	$20 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 50$
Frequency	4	12	18	11	5

Calculate an estimate for the mean height. Give your answer to 1 decimal place.

(4 marks)

8. The price of a laptop, including 20% VAT, is £648. Work out the price of the laptop before VAT was added.

(3 marks)

9. The two-way table shows how the 100 students in Year 11 travel to school.

	Bus	Car	Walk	Total
Boys	18	12	15	45
Girls	22	8	25	55
Total	40	20	40	100

(a) One student is chosen at random. Find the probability that the student walks to school.

(1 marks)

(b) One of the boys is chosen at random. Find the probability that this boy travels by bus.

(2 marks)

(c) Two students are chosen at random from the whole group (without replacement). Find the probability that both are girls who walk to school.

(3 marks)

10. Factorise fully $12x^2 - 4x - 21$.

(3 marks)

11. Two lines L_1 and L_2 intersect at the point $(4, 5)$. L_1 has gradient 2 and L_2 has gradient -1 .

(a) Find the equation of L_1 .

(2 marks)

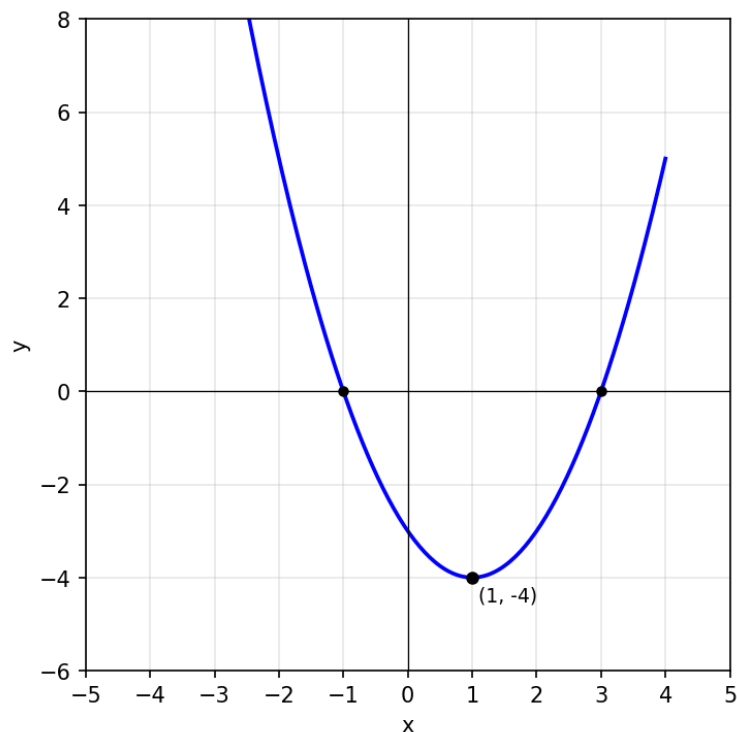
(b) Find the equation of L_2 .

(2 marks)

(c) L_1 crosses the x-axis at A. L_2 crosses the x-axis at B. Work out the length of AB.

(3 marks)

12. The graph shows the curve $y = f(x)$, where $f(x) = x^2 - 2x - 3$. The curve crosses the x-axis at -1 and 3 , and has a turning point at $(1, -4)$.



(a) Write down the coordinates of the turning point of the curve $y = f(x) + 5$.

(1 marks)

(b) Write down the coordinates of the turning point of the curve $y = f(x + 2)$.

(1 marks)

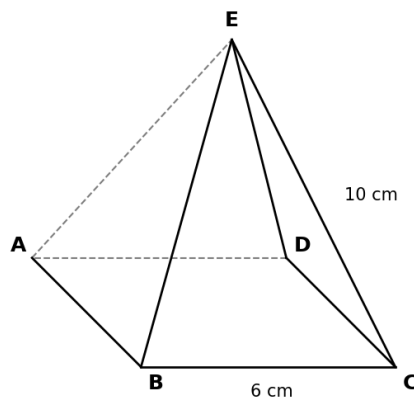
(c) Write down the coordinates of the turning point of the curve $y = -f(x)$.

(1 marks)

13. Simplify fully $(2x^2 + 5x - 12) / (4x^2 - 9)$.

(4 marks)

14. The diagram shows a pyramid with a square base ABCD. Each side of the base is 6 cm, and each slanted edge (EA, EB, EC, ED) is 10 cm. The vertex E is directly above the centre of the base.



(a) Show that the diagonal of the base, AC, is $6\sqrt{2}$ cm.

(2 marks)

(b) Hence work out the vertical height of the pyramid (the distance from E to the base). Give your answer correct to 3 significant figures.

(3 marks)

15. $f(x) = 2x + 7$ and $g(x) = x^2 - 3$.

(a) Find $gf(x)$. Give your answer in expanded form.

(3 marks)

(b) Find $f^{-1}(x)$.

(2 marks)

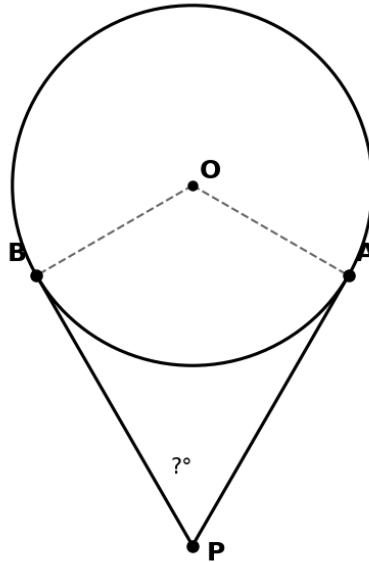
(c) Solve $fg(x) = 19$.

(3 marks)

16. A ball is thrown upwards from a height of 1.5 m. Its height h (in metres) above the ground after t seconds is given by $h = 1.5 + 12t - 5t^2$. Calculate the time at which the ball first hits the ground. Give your answer correct to 2 decimal places.

(4 marks)

17. The diagram shows a circle with centre O . P is a point outside the circle. PA and PB are tangents to the circle, touching it at A and B respectively. The radius of the circle is 2.5 cm and $OP = 5$ cm.



(a) Explain why angle $OAP = 90^\circ$.

(1 marks)

(b) Work out the length of tangent PA . Give your answer in surd form.

(3 marks)

(c) Work out the size of angle APB . Give your answer correct to 1 decimal place.

(3 marks)

18. Sam has £5,000 to invest for 3 years. He is choosing between two accounts:

- **Account A:** 4.5% simple interest per year.
- **Account B:** 4.2% compound interest per year.

Which account would give Sam the larger amount after 3 years, and by how much? Show your working clearly.

(5 marks)

19. $\xi = \{\text{integers from 1 to 20 inclusive}\}$. $A = \{\text{multiples of 3}\}$. $B = \{\text{even numbers}\}$.

(a) List the members of $A \cap B$.

(2 marks)

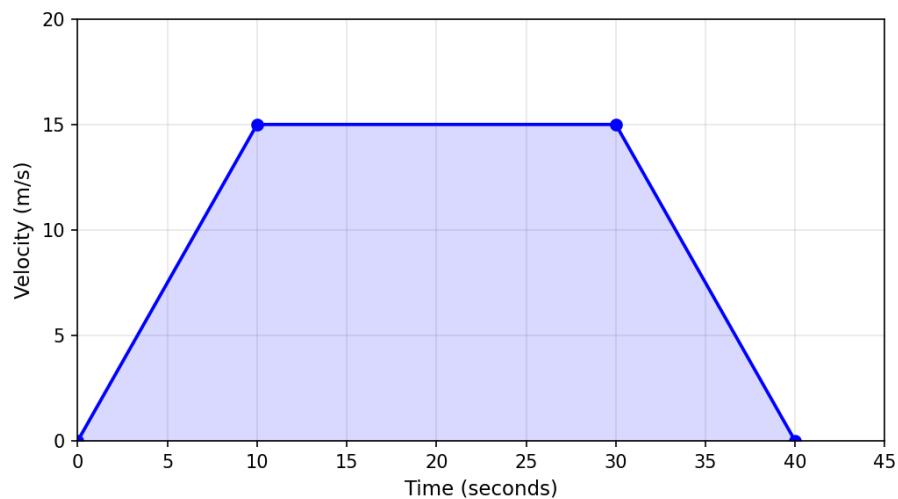
(b) List the members of $A \cup B$.

(2 marks)

(c) A number is chosen at random from ξ . Find the probability that the number is in A but not in B .

(2 marks)

20. The velocity-time graph shows the motion of a car over 40 seconds. The car accelerates uniformly from rest, travels at constant velocity, then decelerates to rest.



(a) Work out the acceleration of the car during the first 10 seconds. State the units.

(2 marks)

(b) Work out the total distance travelled by the car. (This equals the area under the graph.)

(3 marks)

21. Without using a calculator, evaluate:

(a) $27^{2/3}$

(2 marks)

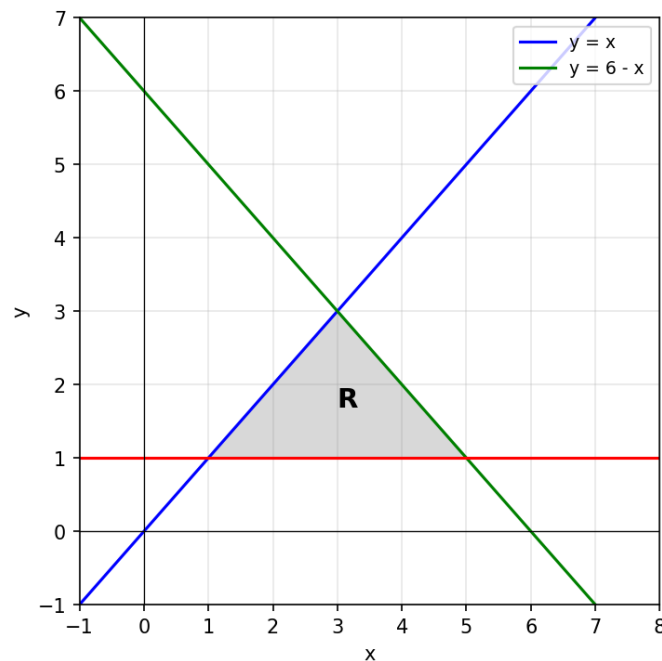
(b) $16^{-3/4}$

(2 marks)

(c) $(8/125)^{2/3}$

(3 marks)

22. The diagram shows three straight lines: $y = x$, $y = 6 - x$, and $y = 1$. The region R is bounded by these three lines and is shaded.



(a) Write down the three inequalities that define the region R.

(3 marks)

(b) List all the points (x, y) inside the region R where both x and y are integers.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 3 Set A

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	£640	M1: 3200×0.04 (= 128 per year). M1: $\times 5$. A1: 640. [3]
2	$3/20$	M1: $(3/8) \times (2/5)$. M1: = 6/40. A1: simplifies to $3/20$. [3]
3	$r \approx 4.92$ cm (3 s.f.)	M1: $500 = (4/3)\pi r^3$. M1: $r^3 = 1500/(4\pi) = 119.37$. A1: $r = 4.92$ (3 s.f.). [3]
4	$n = -1, 0, 1, 2, 3$	M1: solves $-3 < 2n + 1$, giving $n > -2$. M1: solves $2n + 1 \leq 7$, giving $n \leq 3$. A1: lists all integers from -1 to 3 . [3]
5(a)	20 m/s	M1: $72 \text{ km/h} = 72000 \text{ m} / 3600 \text{ s}$. A1: 20 m/s. [2]
5(b)	900 m	M1: 20×45 . A1: 900. [2]
6	66.9° (1 d.p.)	M1: identifies sine rule. M1: $\sin(\text{BCA})/11 = \sin(42^\circ)/8$. M1: $\sin(\text{BCA}) = 11 \times \sin(42^\circ)/8 \approx 0.9197$. A1: angle BCA = $\sin^{-1}(0.9197) = 66.9^\circ$. [4]
7	25.2 cm	M1: midpoints 5, 15, 25, 35, 45. M1: $\Sigma fx = 5(4) + 15(12) + 25(18) + 35(11) + 45(5) = 1260$. M1: $\Sigma f = 50$. A1: $1260/50 = 25.2$. [4]
8	£540	M1: recognises $648 = 120\%$ of original. M1: $648 \div 1.20$. A1: 540. [3]
9(a)	$2/5$	B1: $40/100 = 2/5$. [1]
9(b)	$2/5$	M1: identifies 18 boys take bus, 45 boys total. A1: $18/45 = 2/5$. [2]
9(c)	$2/33$	M1: 25 girls walk; $P(\text{first}) = 25/100$. M1: $P(\text{second} \text{first}) = 24/99$. A1: $(25/100) \times (24/99) = 600/9900 = 2/33$. [3]
10	$(2x + 3)(6x - 7)$	M1: factors of $12 \times (-21) = -252$ summing to -4 : +18 and -14 . M1: splits: $12x^2 + 18x - 14x - 21$. A1: $(2x + 3)(6x - 7)$. [3]
11(a)	$y = 2x - 3$	M1: uses point with gradient: $5 = 2(4) + c$. A1: $y = 2x - 3$. [2]
11(b)	$y = -x + 9$	M1: $5 = -(4) + c$. A1: $y = -x + 9$. [2]
11(c)	AB = 7.5 units	M1: A = (1.5, 0) where $0 = 2x - 3$. M1: B = (9, 0) where $0 = -x + 9$. A1: AB = $9 - 1.5 = 7.5$. [3]
12(a)	(1, 1)	B1: translates up 5. [1]
12(b)	(-1, -4)	B1: translates left 2. [1]
12(c)	(1, 4)	B1: reflects in x-axis. [1]
13	$(x + 4) / (2x + 3)$	M1: factorises numerator: $(2x - 3)(x + 4)$. M1: factorises denominator: $(2x - 3)(2x + 3)$. M1: cancels common factor. A1: $(x + 4)/(2x + 3)$. [4]
14(a)	Shown: $AC = 6\sqrt{2}$	M1: Pythagoras $AC^2 = 6^2 + 6^2 = 72$. A1: $\sqrt{72} = 6\sqrt{2}$. [2]
14(b)	Height = $\sqrt{82} \approx 9.06$ cm (3 s.f.)	M1: distance from centre of base to A = $(1/2)(6\sqrt{2}) = 3\sqrt{2}$. M1: Pythagoras: $EO^2 + (3\sqrt{2})^2 = 10^2$. M1: $EO^2 = 100 - 18 = 82$. A1: $EO = \sqrt{82} \approx 9.06$ cm (3 s.f.). [3]
15(a)	$gf(x) = 4x^2 + 28x + 46$	M1: $g(2x + 7)$. M1: expands $(2x + 7)^2 = 4x^2 + 28x + 49$. A1: $-3 = 4x^2 + 28x + 46$. [3]
15(b)	$f^{-1}(x) = (x - 7)/2$	M1: rearranges $y = 2x + 7$. A1: $(x - 7)/2$. [2]
15(c)	$x = \pm 3$	M1: $fg(x) = 2(x^2 - 3) + 7 = 2x^2 + 1$. M1: $2x^2 + 1 = 19$, $x^2 = 9$. A1: $x = \pm 3$. [3]
16	$t \approx 2.52$ s (2 d.p.)	M1: sets $h = 0$: $5t^2 - 12t - 1.5 = 0$. M1: applies quadratic formula with $a = 5$, $b = -12$, $c = -1.5$. M1: discriminant = $144 + 30 = 174$; $t = (12 \pm \sqrt{174})/10$. A1: positive root $t \approx 2.52$. [4]
17(a)	Tangent $\perp$ radius	B1: states that a tangent to a circle meets the radius at the point of contact at 90° . [1]

Q	Answer	Mark scheme
17(b)	$PA = (5\sqrt{3})/2$ cm	M1: identifies right angle at A. M1: $OP^2 = OA^2 + PA^2$, so $25 = 6.25 + PA^2$. A1: $PA = \sqrt{18.75} = (5\sqrt{3})/2$ (accept $\sqrt{(75/4)}$ or 4.33 cm). [3]
17(c)	60°	M1: in triangle OAP, $\sin(\text{angle APO}) = OA/OP = 0.5$. M1: angle APO = 30° . By symmetry, angle APB = 2×30 . A1: 60.0° . [3]
18	Account A by £18.64	M1: Account A interest = $5000 \times 0.045 \times 3 = 675$. Final = 5675. M1: Account B final = $5000 \times (1.042)^3$. M1: = 5656.36 (to nearest penny). M1: compares: $5675 - 5656.36 = 18.64$. A1: states Account A gives more by £18.64 (accept £18 to £19). [5]
19(a)	{6, 12, 18}	M1: identifies multiples of 6 in ξ . A1: complete list. [2]
19(b)	{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20}	M1: combines multiples of 3 with even numbers. A1: complete list. [2]
19(c)	3/20	M1: A only (not in B) = {3, 9, 15} (multiples of 3 that are odd). A1: 3/20. [2]
20(a)	1.5 m/s^2	M1: gradient = $(15 - 0)/(10 - 0)$. A1: 1.5 m/s^2 (with units). [2]
20(b)	450 m	M1: trapezium with parallel sides 20 and 40, height 15. M1: $(1/2)(20 + 40)(15)$. A1: 450 m. [3]
21(a)	9	M1: $27^{1/3} = 3$. A1: $3^2 = 9$. [2]
21(b)	1/8	M1: $16^{1/4} = 2$; $2^3 = 8$. A1: 1/8 (reciprocal). [2]
21(c)	4/25	M1: $(8/125)^{1/3} = 2/5$. M1: squares. A1: 4/25. [3]
22(a)	$y \leq x$; $y \leq 6 - x$; $y \geq 1$	B1 for each inequality. [3]
22(b)	9 integer points	M1: identifies candidate integer points within boundaries. M1: checks each against all three inequalities. A1: (1,1), (2,1), (3,1), (4,1), (5,1), (2,2), (3,2), (4,2), (3,3). [3]

Worked Solutions — Predicted Paper 3 Set A

Higher Tier · Edexcel 1MA1 style

Q1: Simple interest

Simple interest = principal $\times$ rate $\times$ time.
 $= 3200 \times 0.04 \times 5 = \textbf{£640}.$

Q2: Fractions in context

Fraction who wear both = $(3/8) \times (2/5) = 6/40 = \textbf{3/20}.$

Q3: Sphere radius

$V = (4/3)\pi r^3 = 500.$
 $r^3 = 3 \times 500 / (4\pi) = 1500 / 12.566 = 119.37.$
 $r = (119.37)^{1/3} = \textbf{4.92 cm (3 s.f.)}.$

Q4: Combined inequality

Left part: $-3 < 2n + 1 \Rightarrow -4 < 2n \Rightarrow n > -2.$
Right part: $2n + 1 \leq 7 \Rightarrow n \leq 3.$
Combined: $-2 < n \leq 3.$ Integer values: **$n = -1, 0, 1, 2, 3.$**

Q5: Speed conversion

(a) $72 \text{ km/h} = 72000 \text{ m} / 3600 \text{ s} = \textbf{20 m/s}.$
(b) Distance = speed $\times$ time = $20 \times 45 = \textbf{900 m}.$

Q6: Sine rule

$\sin(\text{BCA}) / AB = \sin(\text{BAC}) / BC.$
 $\sin(\text{BCA}) / 11 = \sin(42^\circ) / 8.$
 $\sin(\text{BCA}) = 11 \times \sin(42^\circ) / 8 = 11 \times 0.6691 / 8 = 0.9200.$
angle BCA = $\sin^{-1}(0.9200) = \textbf{66.9^\circ (1 d.p.)}.$

Q7: Mean from grouped data

Midpoints: 5, 15, 25, 35, 45.
 $\Sigma fx = 5(4) + 15(12) + 25(18) + 35(11) + 45(5) = 20 + 180 + 450 + 385 + 225 = 1260.$
 $\Sigma f = 50.$
Mean = $1260 / 50 = \textbf{25.2 cm}.$

Q8: Reverse percentage

The price includes 20% VAT, so 648 represents 120% of the original.
Original = $648 / 1.20 = \textbf{£540}.$
Check: $540 \times 1.20 = 648. \checkmark$

Q9: Two-way table

(a) Walk total = 40. $P(\text{walks}) = 40/100 = 2/5$.

(b) Boys taking bus = 18 out of 45 boys total. $P = 18/45 = 2/5$.

(c) Girls who walk = 25. $P(\text{first}) = 25/100$.

After taking one such girl, $P(\text{second}) = 24/99$.

Combined = $(25/100) \times (24/99) = 600/9900 = 2/33$.

Q10: Quadratic factorising

Find factors of $12 \times (-21) = -252$ summing to -4 : $+18$ and -14 .

$$12x^2 + 18x - 14x - 21 = 6x(2x + 3) - 7(2x + 3) = (2x + 3)(6x - 7).$$

Q11: Lines through a point

(a) L_1 : $y = 2x + c$. Sub $(4, 5)$: $5 = 8 + c$, so $c = -3$. **$y = 2x - 3$.**

(b) L_2 : $y = -x + c$. Sub $(4, 5)$: $5 = -4 + c$, so $c = 9$. **$y = -x + 9$.**

(c) A on x-axis: $0 = 2x - 3$, $x = 1.5$. So $A = (1.5, 0)$.

B on x-axis: $0 = -x + 9$, $x = 9$. So $B = (9, 0)$.

$AB = 9 - 1.5 = 7.5$ units.

Q12: Graph transformations

Original turning point: $(1, -4)$.

(a) $y = f(x) + 5$: translate up by 5 $\Rightarrow$ **$(1, 1)$.**

(b) $y = f(x + 2)$: translate left by 2 $\Rightarrow$ **$(-1, -4)$.**

(c) $y = -f(x)$: reflect in x-axis $\Rightarrow$ **$(1, 4)$.**

Q13: Algebraic fractions

Numerator: $2x^2 + 5x - 12$. Factors of $2 \times (-12) = -24$ summing to 5: $+8$ and -3 .

$$2x^2 + 8x - 3x - 12 = 2x(x + 4) - 3(x + 4) = (2x - 3)(x + 4).$$

Denominator (difference of squares): $4x^2 - 9 = (2x - 3)(2x + 3)$.

Cancel $(2x - 3)$: answer = **$(x + 4) / (2x + 3)$.**

Q14: Square-based pyramid

(a) By Pythagoras on the square base: $AC^2 = 6^2 + 6^2 = 72$.

$$AC = \sqrt{72} = 6\sqrt{2} \text{ cm. } \checkmark$$

(b) Let O be the centre of the base. $AO = (1/2)AC = 3\sqrt{2}$ cm.

Triangle EAO is right-angled at O (since E is directly above O).

$$EA^2 = EO^2 + AO^2 \Rightarrow EO^2 = 10^2 - (3\sqrt{2})^2 = 100 - 18 = 82.$$

$$EO = \sqrt{82} = 9.06 \text{ cm (3 s.f.)}$$

Q15: Functions

(a) $gf(x) = g(f(x)) = g(2x + 7) = (2x + 7)^2 - 3$.

Expand: $(2x + 7)^2 = 4x^2 + 28x + 49$.

$$gf(x) = 4x^2 + 28x + 46.$$

(b) Let $y = 2x + 7$. Solve for x : $x = (y - 7)/2$. $f^{-1}(x) = (x - 7)/2$.

(c) $fg(x) = f(g(x)) = f(x^2 - 3) = 2(x^2 - 3) + 7 = 2x^2 + 1$.

Set equal to 19: $2x^2 = 18 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$.

Q16: Quadratic in context

Set $h = 0$: $1.5 + 12t - 5t^2 = 0$, or $5t^2 - 12t - 1.5 = 0$.

Quadratic formula: $a = 5$, $b = -12$, $c = -1.5$.

Discriminant = $144 + 30 = 174$.

$t = (12 \pm \sqrt{174}) / 10 = (12 \pm 13.19) / 10$.

Positive root: $t = 25.19 / 10 = \mathbf{2.52 \text{ seconds (2 d.p.)}}$.

Q17: Circle tangents

(a) A tangent to a circle is perpendicular to the radius at the point of contact. OA is the radius and PA is the tangent, so **angle OAP = 90°**.

(b) Right triangle OAP, with hypotenuse OP = 5 and OA = 2.5.

$$PA^2 = OP^2 - OA^2 = 25 - 6.25 = 18.75.$$

$$PA = \sqrt{18.75} = \sqrt{(75/4)} = (\sqrt{75})/2 = \mathbf{(5\sqrt{3})/2 \text{ cm}} \approx 4.33 \text{ cm}.$$

(c) In right triangle OAP: $\sin(\text{angle APO}) = OA/OP = 0.5$.

angle APO = 30°. By symmetry, OPB is also 30°.

Angle APB = 30 + 30 = **60.0°**.

Q18: Simple vs compound interest

Account A (simple): interest = $5000 \times 0.045 \times 3 = \pounds 675$.

Final balance = $5000 + 675 = \mathbf{\pounds 5,675.00}$.

Account B (compound): balance = $5000 \times (1.042)^3$.

$$(1.042)^3 = 1.131272. \text{ Balance} = 5000 \times 1.131272 = \mathbf{\pounds 5,656.36}.$$

Comparison: Account A is higher by $5675 - 5656.36 = \mathbf{\pounds 18.64}$.

Account A is the better choice for this 3-year horizon.

Q19: Set notation

A = multiples of 3 from 1 to 20 = {3, 6, 9, 12, 15, 18}.

B = even numbers from 1 to 20 = {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}.

(a) $A \cap B$ = multiples of 6: **{6, 12, 18}**.

(b) $A \cup B$ = **{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20}**.

(c) A but not B = odd multiples of 3 = {3, 9, 15}. Probability = **3/20**.

Q20: Velocity-time graph

(a) Acceleration = gradient of the first section = $(15 - 0)/(10 - 0) = \mathbf{1.5 \text{ m/s}^2}$.

(b) Distance = area under the graph (a trapezium).

Parallel sides: constant-velocity section (20 s) and total time (40 s). Height: 15 m/s.

$$\text{Area} = (1/2)(20 + 40)(15) = \mathbf{450 \text{ m}}.$$

Q21: Fractional and negative indices

(a) $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$.

(b) $16^{-3/4} = 1 / 16^{3/4} = 1 / (16^{1/4})^3 = 1 / 8 = 1/8$.

(c) $(8/125)^{2/3} = ((8/125)^{1/3})^2 = (2/5)^2 = 4/25$.

Q22: Region of inequalities

(a) R is below $y = x$, below $y = 6 - x$, and above $y = 1$.

Inequalities: $y \leq x$, $y \leq 6 - x$, $y \geq 1$.

(b) Integer points satisfying all three:

$y = 1$: need $1 \leq x$ and $1 \leq 6 - x$ (i.e. $x \leq 5$). Points: (1,1), (2,1), (3,1), (4,1), (5,1).

$y = 2$: need $2 \leq x \leq 4$. Points: (2,2), (3,2), (4,2).

$y = 3$: need $3 \leq x \leq 3$. Point: (3,3).

Total 9 points: (1,1), (2,1), (3,1), (4,1), (5,1), (2,2), (3,2), (4,2), (3,3).